

AI-BLOGNEST MASTER DOCUMENTATION

SOFTWARE ENGINEERING & AI IMPLEMENTATION REPORT

Team ID: SWTID-2026-3083

Project Name: AI-BLOGNEST

Team Leader: MUTHU JEEVANANTHAM M

Team Members: NAGARAJ K | NAVEEN RAJ S | PINKU H

DOCUMENT 1: PROJECT PROPOSAL AND SYNOPSIS

1. Executive Summary & Problem Statement

AI-BLOGNEST is an advanced full-stack blogging platform built with React, Node.js, Express, MongoDB, and integrated Generative AI capabilities. Traditional blogging platforms lack real-time AI assistance for content creation, SEO optimization, auto-tagging, and automated summary generation. AI-BLOGNEST solves these challenges by providing authors and content creators with an intelligent environment that enhances productivity and engagement.

2. Project Vision & Objectives

Vision: To empower content creators with AI-driven writing assistance, automatic SEO formatting, and seamless social publishing.

Objectives:

- Provide secure multi-role authentication (User, Author, Admin).
- Enable AI post generation and automated article summarization.
- Deliver high-performance RESTful API endpoints for blog management.
- Ensure robust data persistence using MongoDB.

DOCUMENT 2: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

1. Hardware Requirements

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	8 GB Minimum (16 GB Recommended)
Storage	256 GB SSD

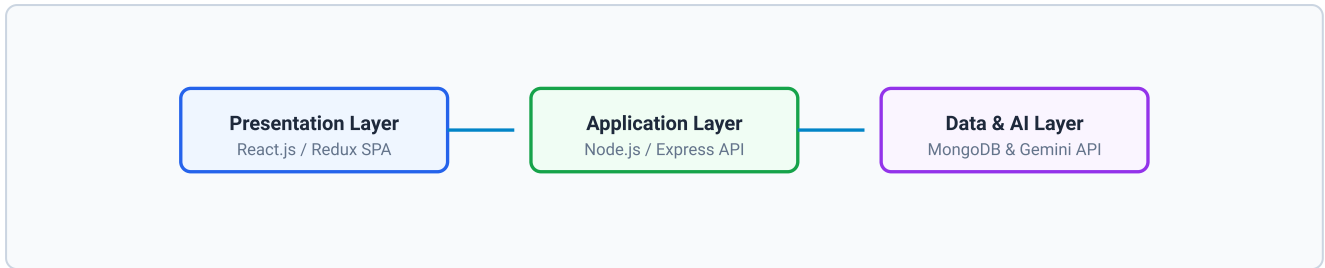
2. Software & Technology Stack

Layer	Technology
Frontend	React.js, Redux Toolkit, Tailwind CSS, Vite
Backend	Node.js, Express.js Framework

DOCUMENT 3: SYSTEM ARCHITECTURE DOCUMENT

1. Architectural Overview

AI-BLOGNEST follows a modern 3-Tier Client-Server Architecture consisting of the Presentation Layer (React SPA), Application Layer (Express REST API), and Data Layer (MongoDB).



2. High-Level Component Design

Module	Description
Auth Module	JWT Authentication, Password Hashing (bcrypt), Role Middleware
Blog Engine	CRUD operations for posts, categories, tags, and comments
AI Module	Service layer communicating with LLM APIs for generation & summary

DOCUMENT 4: DETAILED DESIGN DOCUMENT

1. Detailed Design Specification

This document details the object model, class structures, database schemas, and API design specifications for the AI-BLOGNEST project.

2. Database Schema Design (MongoDB)

Collection	Fields	Relationships
Users	id, name, email, password, role, createdAt	One-to-Many with Posts
Posts	id, title, content, summary, authorId, tags, status	Belongs to User, Has Many Comments
Comments	id, postId, userId, content, createdAt	Belongs to User and Post

DOCUMENT 5: FRONTEND DESIGN AND IMPLEMENTATION DOCUMENT

1. Frontend Architecture (React & Redux)

The frontend is structured with component modularity. Features include responsive UI layouts, rich text editing, state management via Redux Toolkit, and asynchronous API calls using Axios.

2. Key UI Modules

- Home & Feed:** Infinite scroll blog listing with category filtering.
- Post Editor:** Rich text editor with built-in AI completion prompt button.
- Dashboard:** Analytics on post views, user engagements, and published articles.

DOCUMENT 6: BACKEND AND API DOCUMENTATION

1. Backend & REST API Specification

The backend service exposes secure REST endpoints built using Express.js middleware for authentication, input validation, and error handling.

2. Core API Endpoints

Endpoint	Method	Description
/api/auth/register	POST	User registration
/api/auth/login	POST	User login & JWT issuance
/api/blogs	GET / POST	Fetch all posts or create a new post
/api/ai/generate	POST	Generate blog content via AI API

DOCUMENT 7: AI INTEGRATION AND FEATURE SPECIFICATION

1. Artificial Intelligence Integration

AI-BLOGNEST incorporates Large Language Model (LLM) APIs to assist users in blog content generation, title suggestion, grammar improvement, and auto-summarization.

2. AI Service Workflow

1. User submits topic prompt in Post Editor.
2. Backend AI controller structures prompt with system context.
3. Request sent to LLM service.
4. Output formatted and stream-returned to client editor UI.

DOCUMENT 8: TEST PLAN AND TEST CASES DOCUMENT

1. Software Testing Plan & Results

Comprehensive testing was conducted across unit, integration, and end-to-end user acceptance levels.

2. Test Cases & Execution Matrix

Test ID	Test Scenario	Expected Result	Status
TC-001	User Login with valid credentials	JWT Token returned, Redirected	PASS
TC-002	AI Blog Generation Prompt	AI text populated in editor	PASS
TC-003	Create Post without Title	Validation error message shown	PASS

DOCUMENT 9: USER MANUAL AND DEPLOYMENT GUIDE

1. Installation Instructions

```
1. Clone repository: git clone https://github.com/muthujeeva522-gif/Blognest.git
2. Install Backend Dependencies: cd server && npm install
3. Install Frontend Dependencies: cd client && npm install
4. Configure environment variables (.env)
5. Start development servers: npm run dev
```

DOCUMENT 10: FINAL PROJECT REPORT AND CONTRIBUTION SUMMARY

1. Final Project Report & Conclusion

The AI-BLOGNEST project was successfully developed, tested, and documented by Team SWTID-2026-3083. The platform seamlessly combines modern web engineering practices with cutting-edge AI assistance.

2. Team Contribution Matrix

Team Member	Role & Key Responsibilities
MUTHU JEEVANANTHAM M (Leader)	Architecture Design, Backend API & AI Integration
NAGARAJ K	Frontend React Component Development & UI/UX Design
NAVEEN RAJ S	MongoDB Database Schema, State Management & Testing
PINKU H	Authentication, Security Middleware & Documentation